

Econ 0100 | Classwork E3

Due in Recitation

Question 1 (of 1) | Big Flue Networks

Flue networks forms a convenient form of communication and transportation for those in the wizarding world. Its easy to move between flue portals on the network or use it to chat with those at a distance. Due to the benefits of adding *to* an existing network instead of starting a new one (i.e. network effects) there have historically only been two networks: Fast Flue Inc. and Soot Net. Suppose both networks have a constant marginal cost of 10 galleons, no fixed costs, and face a demand curve for flue network trips given by the following.

$$P = 200 - Q$$

The marginal revenue for Fast Flue Inc. is the following.

$$MR_F = 200 - 2q_F - q_S$$

The marginal revenue for Soot Net. is the following.

$$MR_S = 200 - 2q_S - q_F$$

The networks use a sophisticated ticketing system to charge a price for every trip. Prices are in galleons. This market is symmetric.

Q1.a | Fast Flue Inc.'s Best Response

What is the best response for Fast Flue Inc.?

$$\begin{aligned} MC &= MR_F \\ 10 &= 200 - 2q_F - q_S \\ 2q_F &= 190 - q_S \\ BR_F: q_F &= 95 - \frac{1}{2}q_S \end{aligned}$$

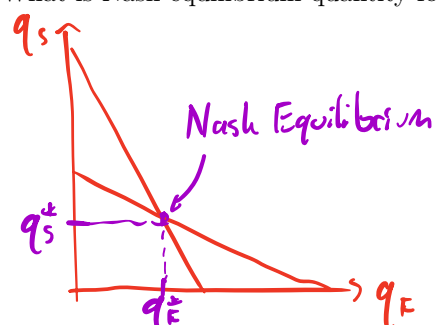
Q1.b | Soot Net's Best Response

What is the best response for Soot Net?

$$\begin{aligned} \text{This game is symmetric. :)} \\ BR_S: q_S &= 95 - \frac{1}{2}q_F \end{aligned}$$

Q1.c | Market Quantity

What is Nash equilibrium quantity for Fast Flue Inc. and Soot Net?



$$q_F^* = 95 - \frac{1}{2} \left(95 - \frac{1}{2} q_F^* \right)$$

$$\frac{3}{4} q_F^* = 95 - \frac{1}{2} 95 = \frac{1}{2} 95$$

→

$$q_F^* = \frac{190}{3}$$

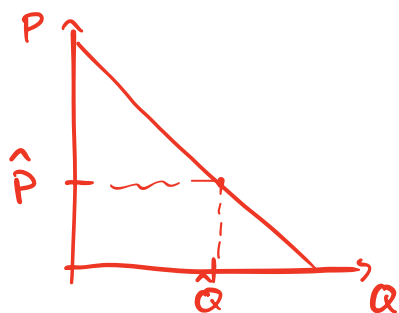
This game is symmetric, so

→

$$q_s^* = \frac{190}{3}$$

Q1.d | Equilibrium Price

What is the Nash equilibrium price in this market?



$$\hat{Q} = q_F^* + q_s^* = \frac{380}{3}$$

$$\hat{P} = 200 - \frac{380}{3} = \frac{220}{3}$$

Q1.e | Market Profit

What is the total profit (sum of both firms profit) in this market?

When MC is constant and there are no fixed costs, $ATC = MC$. So $ATC = 10$.

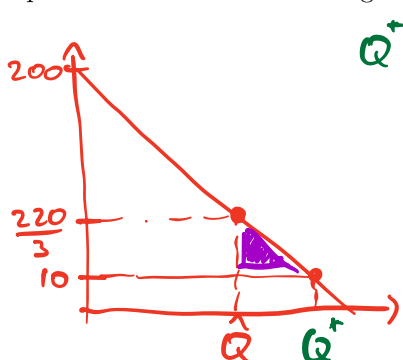
$$\begin{aligned} \pi_F^* &= q_F^* (\hat{P} - ATC^*) \\ &= \frac{190}{3} \left(\frac{220}{3} - 10 \right) \end{aligned}$$

Since the game is symmetric, $\pi_s = \pi_F$.

$$\rightarrow \pi = 2 \cdot \frac{190}{3} \left(\frac{220}{3} - 10 \right) = 2 \cdot \frac{190}{3} \cdot \frac{190}{3}$$

Q1.e | Deadweight Loss

Graph and solve for the deadweight loss.



$$Q^*: 10 = 200 - Q^*$$

$$Q^* = 190$$

$$\begin{aligned} \Delta &= \frac{1}{2} \left(\frac{220}{3} - 10 \right) \left(190 - \frac{380}{3} \right) \\ &= \frac{1}{2} \cdot \frac{190}{3} \cdot \frac{190}{3} \end{aligned}$$